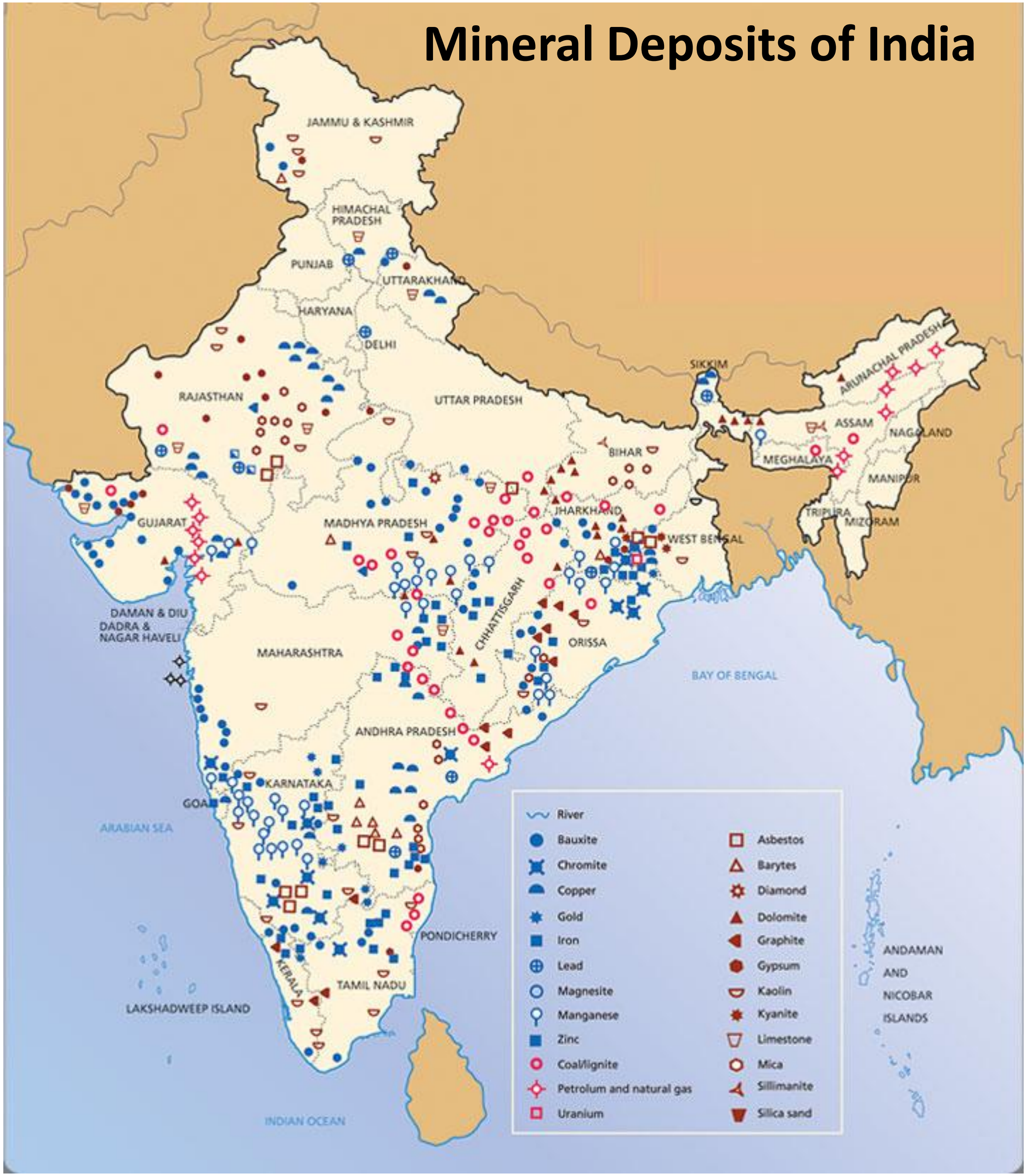
# Mineral Deposits of India



# Indian Distribution of Ore deposits

- **Iron Ore Deposit:** Hematite ($Fe_2O_3$) and Magnetite ($Fe_3O_4$), Goethite [FeO(OH)]
- Hematitic ores are distributed over Jharkhand, Odisha, Chhatisgarh, Goa, Karnataka and Maharashtra.
- Magnetite ores are distributed over Karnataka, Andhra.
- Unlike hematite which is red in colour, magnetite's in India are blackish to brown. Magnetite ores are sometimes associated with quartz schist's and other metamorphosed rocks. Pradesh and Tamil Nadu

# Major Iron Ore Deposits in India

Kiriburu, Chiria, Noamundi, Jamda-Iron ore mines in Jharkhand

Joda East, Bolani, Khondbond, Daitari, Tensa-Iron Ore mines in Odisha

Dalli-Rajhara, Bailadila Iron Ore Mines in Chhatisgarh

Bellary-Hospet, Kudremukh and Chitradurga-Tumkur Iron Ore Deposits in Karnataka

# GOETHITE

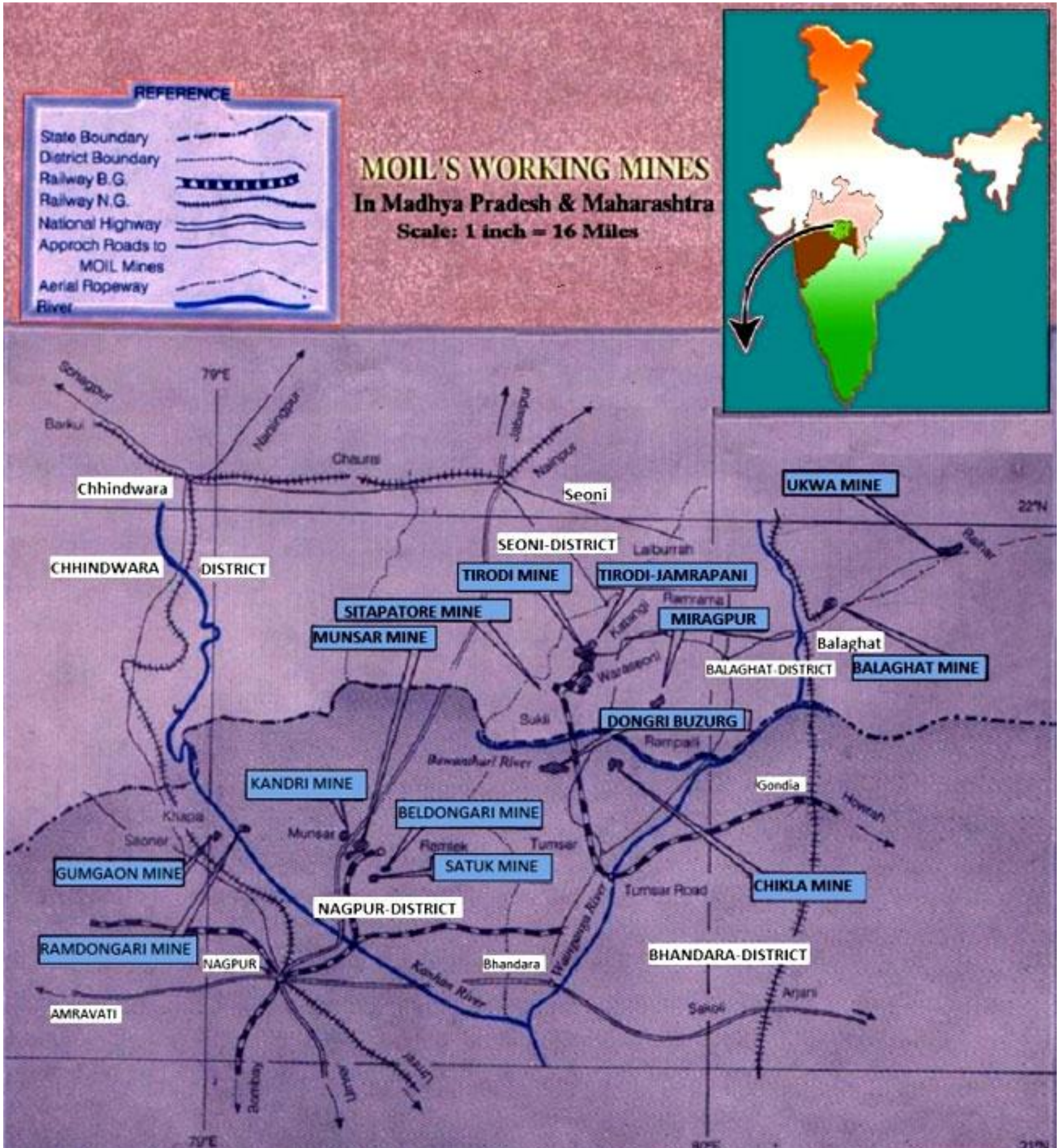
# Manganese

- Odisha, Jharkhand, Madhya Pradesh, Maharashtra, Andhra Pradesh, Goa.
- Like iron-ore, manganese is also extracted in Orissa from Kalahandi, Bolangir, Koraput, and Keonjhar districts
- Madhya Pradesh-located in the districts of Balaghat, Chindwara and Shadol district.
- Important Mn Ore Minerals are pyrolusite, psilomelane, braunite, rhodonite, romanechite, hausmannite

## Chromite ($FeOCr_2O_3$):

Sukinda and Nuasahi area of Odisha, Chitradurga, Hassan, Shimoga districts of Karnataka, Bhandara and Ratnagiri districts of Maharashtra, Singbhum district , Jharkjhand (Roro-Jojohatu)

Distribution of Manganese Ore
www.pmfias.com
www.pmfias.com
Chhindwara
Balaghat
Nagpur
Bhandra
Sundargarh
Sambalpur
Bolangir
Kalahandi
Koraput
Srikakulam
Vishakapattanam
Ratnagiri
Bellary
Guntur
Uttar Kanada
Durg
Cuddapah
Shimoga
Tumkur


MOIL'S WORKING MINES
In Madhya Pradesh & Maharashtra
Scale: 1 inch = 16 Miles
REFERENCE
State Boundary
District Boundary
Railway B.G.
Railway N.G.
National Highway
Approch Roads to MOIL Mines
Aerial Ropeway
River
Chhindwara
CHHINDWARA
DISTRICT
Seoni
SEONI-DISTRICT
UKWA MINE
TIRODI MINE
TIRODI-JAMRAPANI
SITAPATORE MINE
MIRAGPUR
MUNSAR MINE
Balaghat
BALAGHAT-DISTRICT
BALAGHAT MINE
DONGRI BUZURG
KANDRI MINE
BELDONGARI MINE
Gondia
SATUK MINE
GUMGAON MINE
CHIKLA MINE
NAGPUR-DISTRICT
RAMDONGARI MINE
NAGPUR
Bhandara
BHANDARA-DISTRICT
AMRAVATI
22°N
21°N
79°E
80°E

- **Bauxite:** Jharkhand (Lohardaga), Orissa Panchpatmalli Hills, Damanjodi), Chhattisgarh, Madhya Pradesh, Gujarat, Maharashtra (Ratnagiri) and Tamil Nadu (Shevroy Hills) are the states with bauxite deposits.
- **Mica:** Mica is mainly found in Jharkhand, Bihar, Andhra Pradesh and Rajasthan. India is the largest producer and exporter of mica.
- **Diamond:** Associated with Kimberlite pipes and lamproite. Found in the states of Andhra Pradesh (Wjarakarur), Chhatisharh (Mahasamund dist), Madhya Pradesh (Panna area)

- **Copper:** Copper is mainly found in Rajasthan (Khetri), Madhya Pradesh (Malanjkhand in Balaghat district), Jharkhand (Surda, Rakha, Mosabani), Karnataka and Andhra Pradesh.
- Singhbhum Shear Zone (Jharkhand) hosts Cu-U deposits
- The copper ores include **Chalcopyrite, bornite, chalcocite, covelite, native copper, malachite**.

- **Gold:** Gold is found in Kolar, Hutti, Gadag and Ajanahalali in Karnataka.
- The Kolar mines are among the deepest mines in the world.
- Kundarkocha and Lawa gold deposit in Jharkhand.

# Pb and Zn Deposit in India

- **Galena** (PbS) is the ore mineral for Pb &
- **Sphalerite** (ZnS) is the ore mineral for Zn

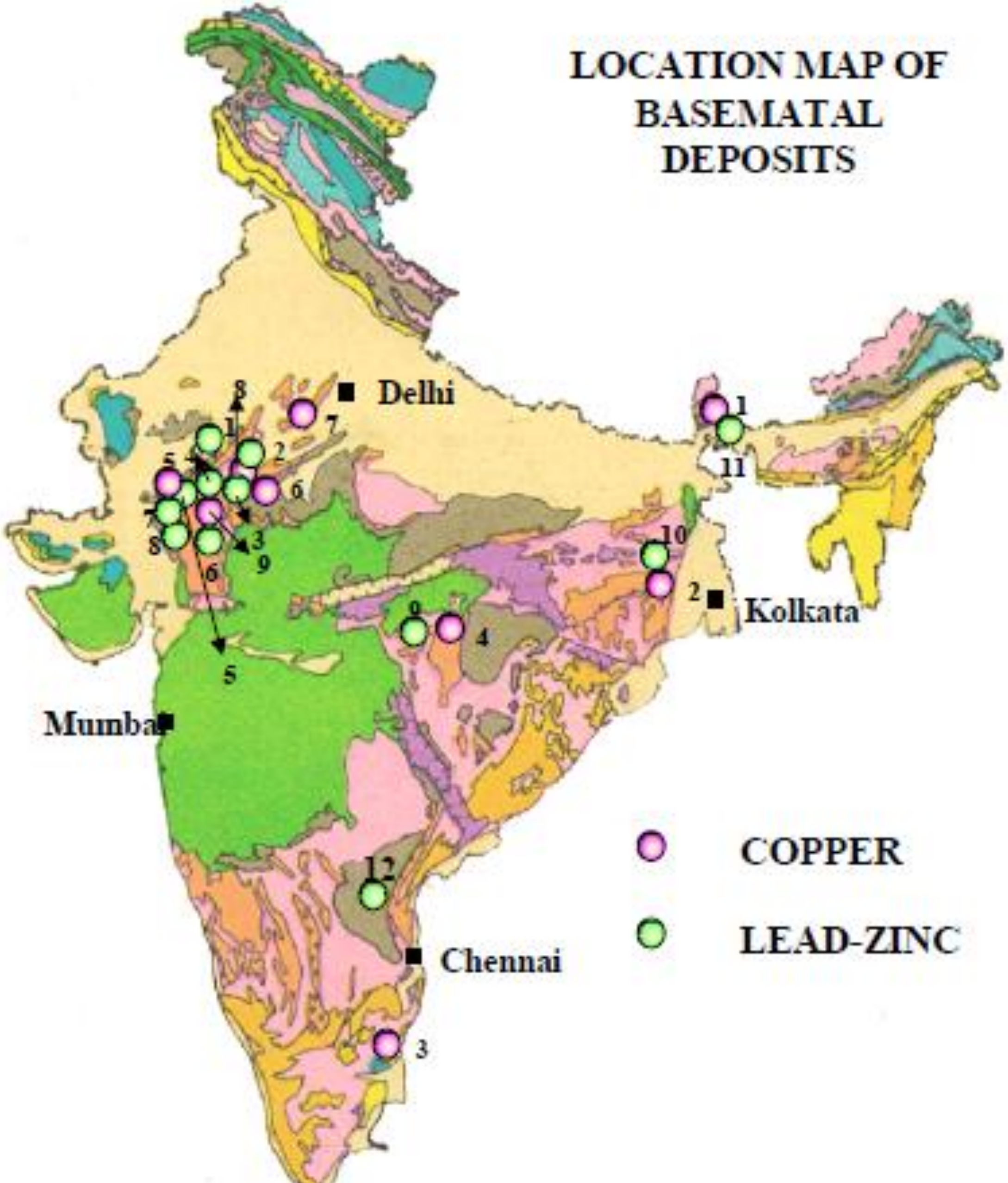
**Pb and Zn deposits:**

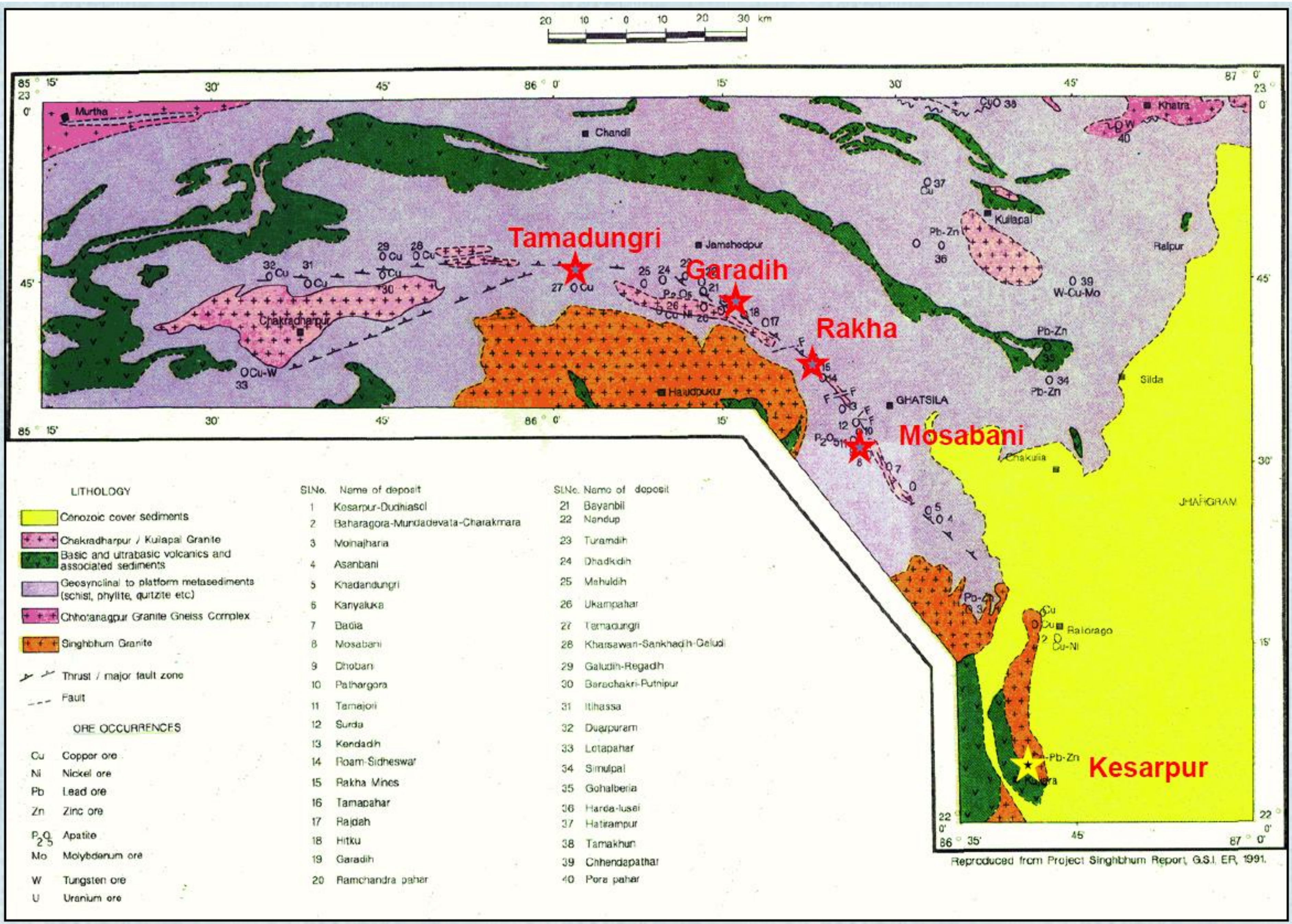
Rampura-Agucha (Bhilwara district), Rajpura Dariba, Sindesar-Khurd (both in Rajsamand district) and Zawar (Udaipur district), Rajasthan.

Sargipalli in Odisha (No mining)

Agnigundala in Andhra Pradesh

# BASE METAL

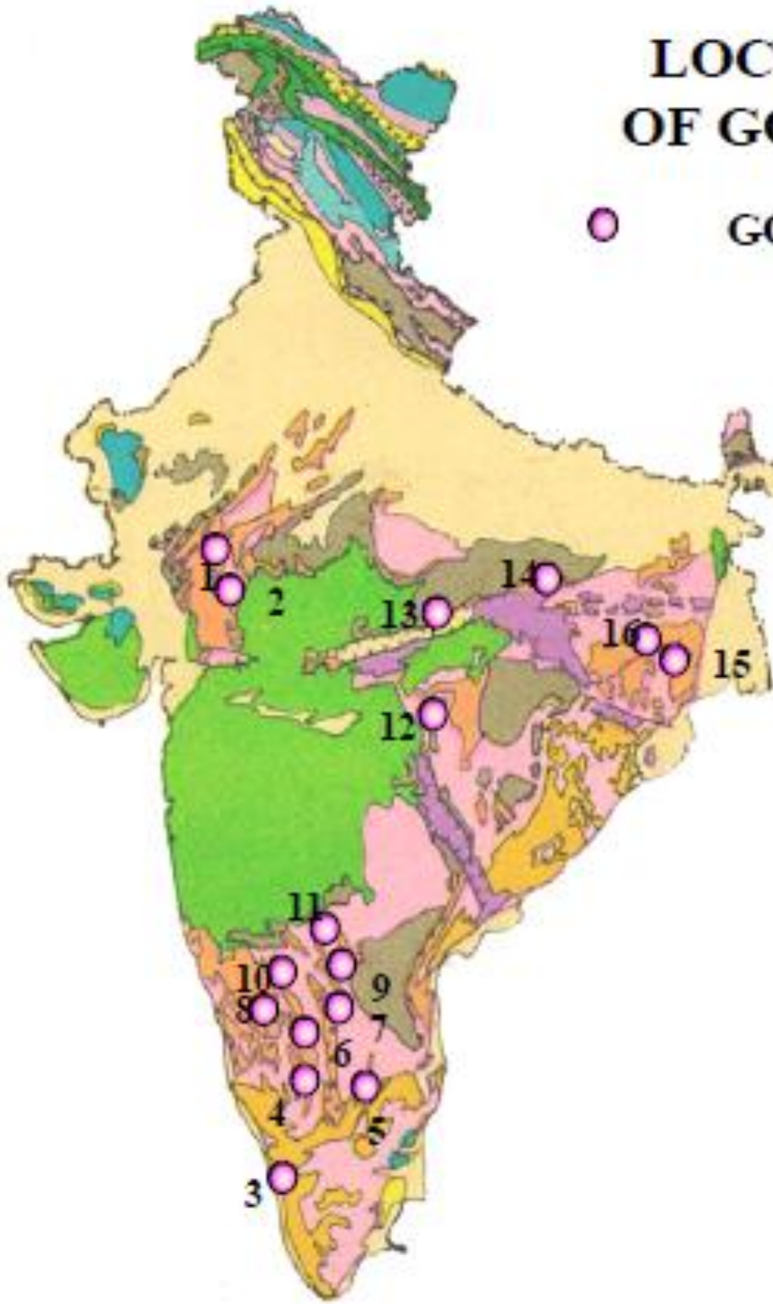

LOCATION MAP OF
BASEMATAL
DEPOSITS
Delhi
Kolkata
Mumbai
Chennai
COPPER
LEAD-ZINC
1
2
3
4
5
6
7
8
9
10
11
12



20 10 0 10 20 30 km
Tamadungri
Garadih
Rakha
Mosabani
Kesarpur
Murtha
Chandil
Jamshedpur
Khatra
Kuilapal
Raipur
Chakradharpur
Haludpukur
GHATSILA
Silda
Chakulia
JHARGRAM
Balorago
LITHOLOGY
Cenozoic cover sediments
Chakradharpur / Kuilapal Granite
Basic and ultrabasic volcanics and associated sediments
Geosynclinal to platform metasediments (schist, phyllite, quitzite etc)
Chhotanagpur Granite Gneiss Complex
Singhbhum Granite
Thrust / major fault zone
Fault
ORE OCCURRENCES
Cu Copper ore
Ni Nickel ore
Pb Lead ore
Zn Zinc ore
P2O5 Apatite
Mo Molybdenum ore
W Tungsten ore
U Uranium ore
Sl.No. Name of deposit
1 Kosarpur-Dudhiasol
2 Baharagora-Mundadevata-Charakmara
3 Moinajharia
4 Asanbani
5 Khadandungri
6 Kanyaluka
7 Badia
8 Mosabani
9 Dhobani
10 Pathargora
11 Tamajori
12 Surda
13 Kendadih
14 Roam-Sidheswar
15 Rakha Mines
16 Tamapahar
17 Rajdah
18 Hitku
19 Garadih
20 Ramchandra pahar
Sl.No. Name of deposit
21 Bayanbil
22 Nandup
23 Turamdih
24 Dhadkidih
25 Mahuldih
26 Ukampahar
27 Tamadungri
28 Kharsawan-Sankhadih-Galudi
29 Galudih-Regadih
30 Barachakri-Putnipur
31 Itihassa
32 Duarpuram
33 Lotapahar
34 Simulpal
35 Gohalberia
36 Harda-lusai
37 Hatirampur
38 Tamakhun
39 Chhendapathar
40 Pora pahar
Reproduced from Project Singhbhum Report, G.S.I. ER, 1991.

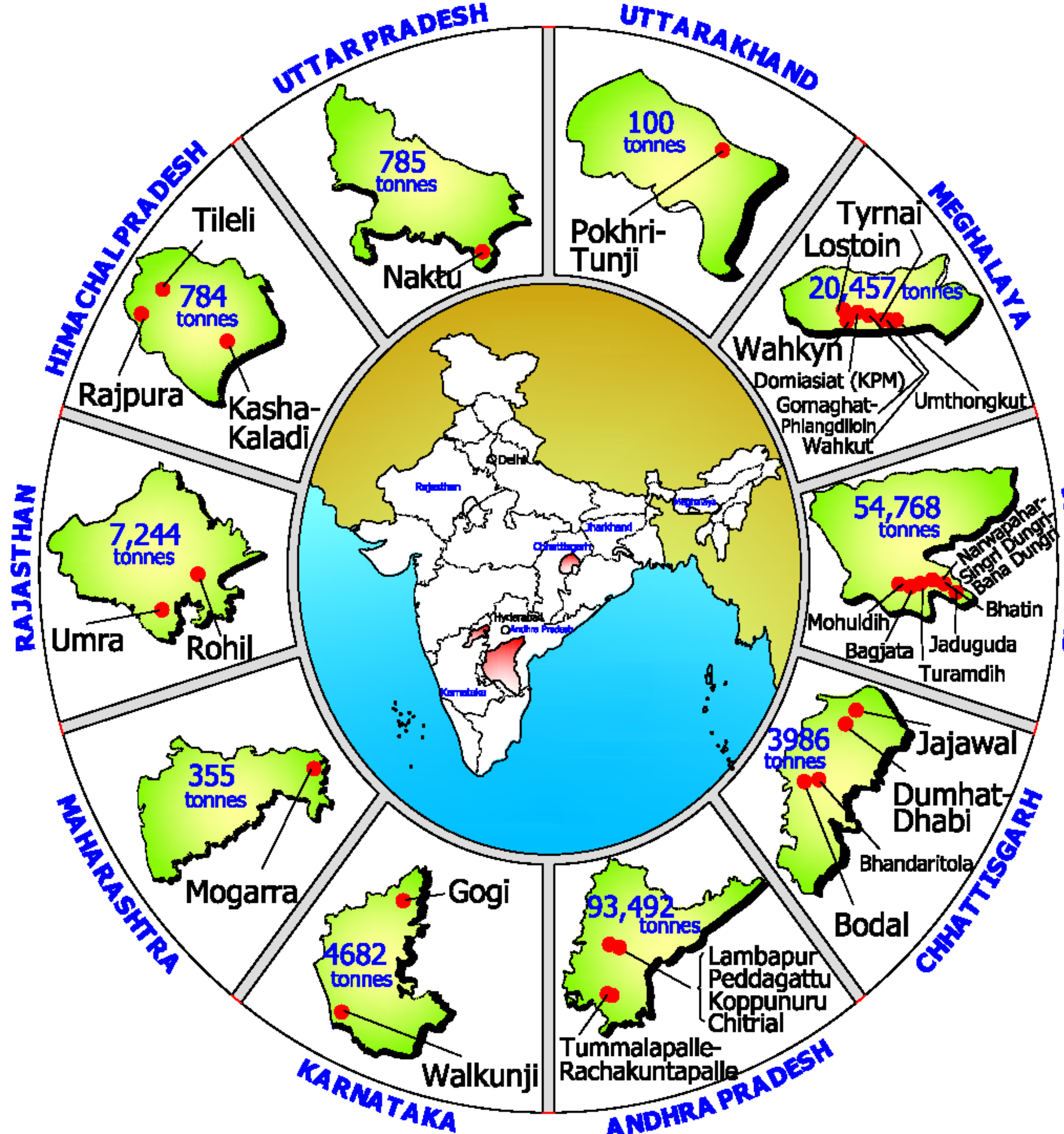
# LOCATION MAP OF GOLD DEPOSITS

GOLD




1. Dugocha
2. Bhukia
3. Kottathara,Puttumala (E)
4. Kempinkote
5. Chigargunta, Mallappakonda
6. Ajjanahalli
7. Bhadrampalle, Ramapuram
8. Chinmulgund
9. Dona East
10. Mysore Mine. Kabuliyatkutti
11. Hutti Mine, Kadoni
12. Parsori,Kitari
13. Imaliya
14. Gurhapahar
15. Kunderkocha
16. Parasi

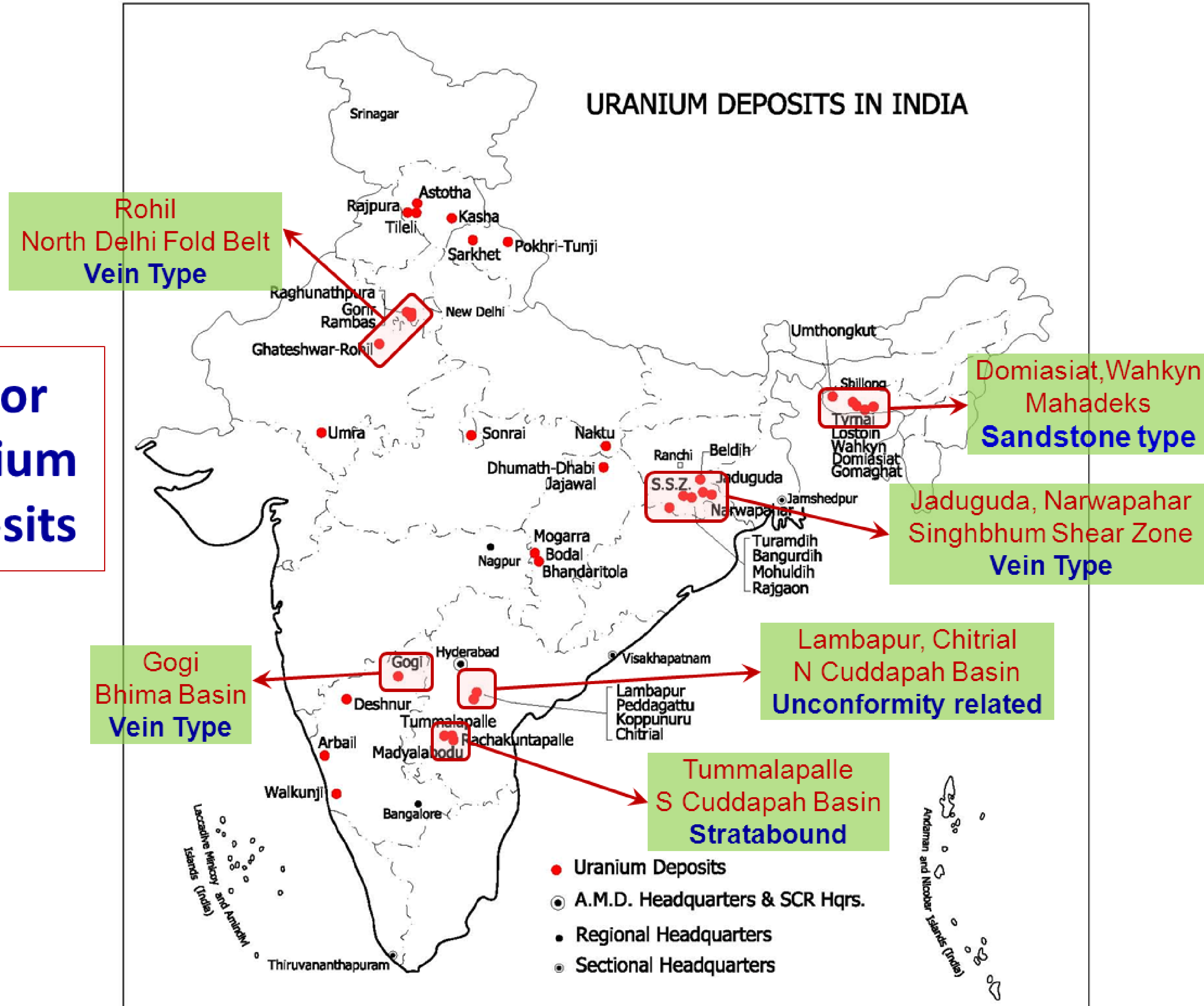
# Distribution of Uranium Reserves

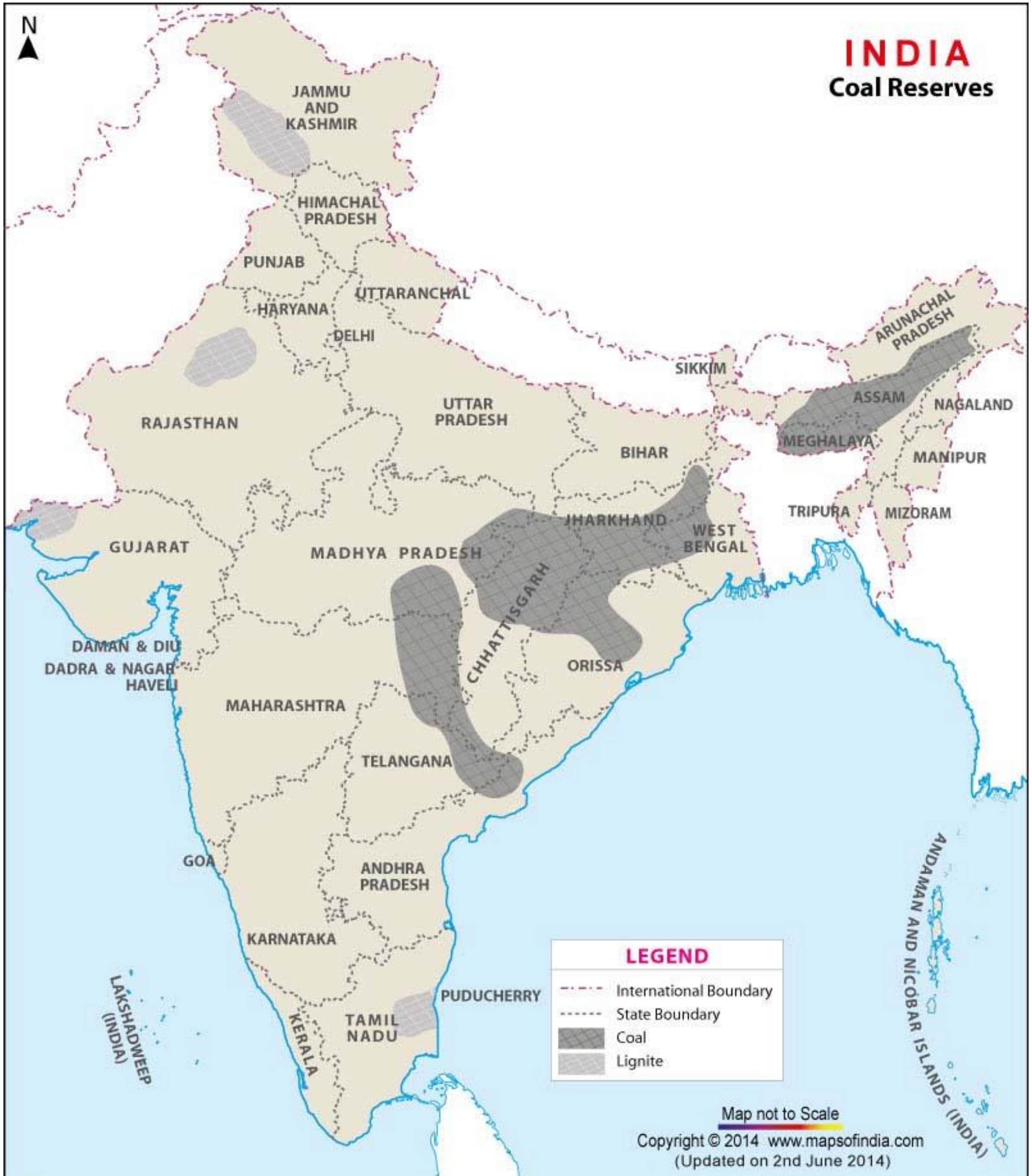



**Total : 1,86,653 t $U_3O_8$**

**(As on October, 2012)**

# Major Uranium Deposits





N
INDIA
Coal Reserves
JAMMU AND KASHMIR
HIMACHAL PRADESH
PUNJAB
UTTARANCHAL
HARYANA
DELHI
RAJASTHAN
UTTAR PRADESH
SIKKIM
ARUNACHAL PRADESH
ASSAM
NAGALAND
MEGHALAYA
MANIPUR
BIHAR
TRIPURA
MIZORAM
JHARKHAND
WEST BENGAL
GUJARAT
MADHYA PRADESH
CHHATTISGARH
ORISSA
DAMAN & DIU
DADRA & NAGAR HAVELI
MAHARASHTRA
TELANGANA
GOA
ANDHRA PRADESH
KARNATAKA
PUDUCHERRY
TAMIL NADU
KERALA
LAKSHADWEEP (INDIA)
ANDAMAN AND NICOBAR ISLANDS (INDIA)
LEGEND
International Boundary
State Boundary
Coal
Lignite
Map not to Scale
Copyright © 2014 www.mapsofindia.com
(Updated on 2nd June 2014)

# INTRODUCTION TO COAL FROMING PROCESS AND DISTRIBUTION

# DEPOSITION OF COAL

- Coal, the most important fossil fuel is a combustible sedimentary rock occuring in workable quantities as seams.
- Coal is formed from accumulation of plant remains modified by biological, physical and chemical processes during and after burial.
- The degree of alteration of the deposits attained determines their rank in the coallfication series from Peat to Anthracite.

# FORMATION OF COAL

- There are two stages of coal formation from plant materials.
- The biochemical period of conversion of plant materials in to peat (Humification).
- The geochemical period of conversion of peat to coal (Coalification)

# HUMIFICATION

- During biochemical period, the changes in the plant debris are brought about mainly by the micro-organisms by decay and decomposition of chemical substances like lignins, celluloses, protiens, waxes and resins.

- The products of such decay is porous, fibrous humic substance called **PEAT**.

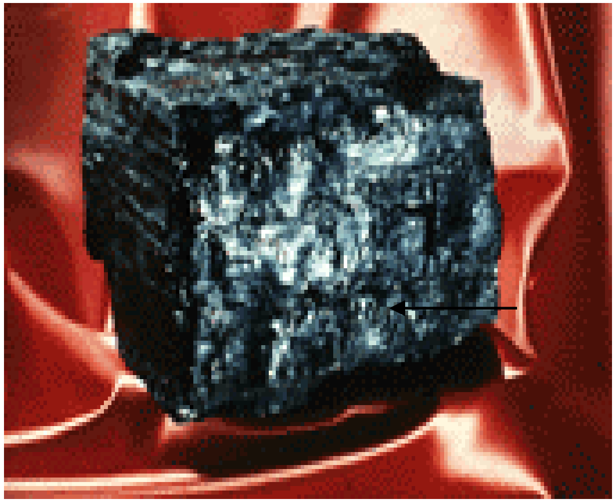
# COALIFICATION

- At this stage $CH_4$, $CO_2$ and $H_2O$ are elimminated and peat is converted into lignite to anthracite depending on the conditions.

- The major reaction in the normal coalification process involves decarboxylation and dehydroxylation leading to the enrichment of carbon.

- The coalification process is primarily governed by rise of temperature during the process.

# COAL CHARACTERISATION



## CHEMICAL ANALYSIS

- PROXIMATE (Moisture, Ash content
- ULTIMATE (C, H, N, S, O)
- SPECIAL TEST

## PETROGRAPHIC ANALYSIS

- MACERAL COMPOSITION
- REFLECTANCE

## WASHABILITY STUDY

- SCREEN ANALYSIS
- FLOAT & SINK TEST

## CBM STUDY

- DESORPTION STUDY
- MEGASCOPIC AND MICROSCOPIC CLEAT STUDY
- GAS COMPOSITION STUDY

## MINERALOGICAL STUDY

- THRU XRD AND SEM

# RANKS OF COAL

- Coal is classified into four main types or ranks (Lignite, Sub-bituminous, bituminous, Anthracite), depending upon the amounts and types of carbon it contains and on the amount of heat energy it can produce.
- The rank of a deposit of coal depends on the pressure and heat acting on the plant debris as it sank deeper and deeper over millions of years.
- For the most part, the higher ranks of coal contain more heat producing energy.

# RANKS OF COAL

- **Lignite** is the lowest rank of coal with the lowest energy content. Lignites tend to be relatively young coal deposits that were not subjected to extreme heat or pressure. Lignite is crumbly and has high moisture content.
- **Sub-bituminous** coal has a higher heating value than lignite. Sub-bituminous coal typically contains 35-45 percent carbon, compared to 25-35 percent for lignite.

# RANKS OF COAL

- **Bituminous** coal contains 45-86 percent carbon, and has two to three times more than the heating value of lignite. Bituminous coal was formed under high heat and pressure.
- Our Singareni coals are mostly Sub-bituminous to bituminous.
- **Anthracite** contains 86-97 percent carbon and its heating value is slightly higher than bituminous coal.

# USGS Classification

Peat.—Contains approximately 85% moisture, 10·4% volatile matter, 4·6% fixed carbon; 1,290 B.T.U.

Lignite.—Brown, woody or composed of finely divided plant tissues, or amorphous and representing the first stage in the development of cannel. Contains 25 to 45% moisture; on drying shrinks and breaks up in an irregular manner; 6,000 to 7,500 B.T.U.

Sub-bituminous.—Black in colour; 12 to 25% moisture; slacks on exposure; 7,000 to 11,000 B.T.U.

Bituminous.—Slacks little on exposure; 11,000 to 15,000 B.T.U. Fuel ratio below 2·5; this class includes cannels, some of the best steam coals and the best gas and by-product coals.

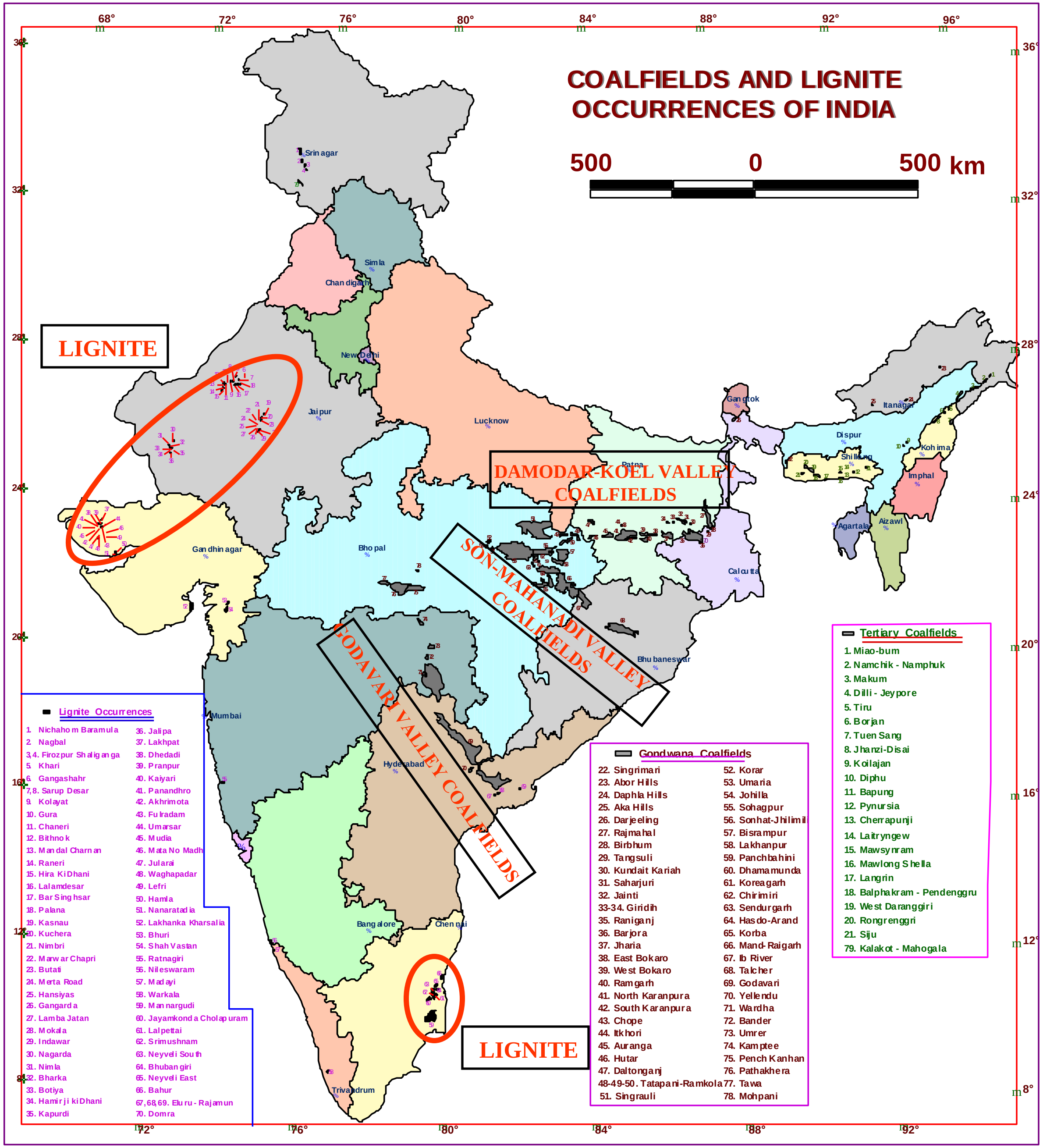
Semi-bituminous.—Nearly smokeless, usually friable and slacks easily; 12,000 to 15,400 B.T.U. Fuel ratio 2·5 to 5.

Semi-anthracite.—Harder than bituminous; burns with a short yellow flame at first and then with a blue flame; fuel ratio 5 to 10.

Anthracite.—Hard, burns with a blue flame; does not soil the hands, difficult to ignite but burns without smoke. Fuel ratio over 10.

# SEYLER'S CLASSIFICATION OF COAL

| | Anthracitic. | Carbonaceous. | Bituminous. | | | Lignitous. | |
|---|---|---|---|---|---|---|---|
| | | | Meta- | Ortho- | Para- | Meta- | Ortho- |
| Genus .. | C over 93·3 | 93·3 to 91·2 | 91·2 to 89·0 | 89·0 to 87·0 | 87·0 to 84·0 | 84 to 80 | 80 to 75 |
| Per-bituminous | .... | .... | Per-Meta-Bituminous. Northern Steam Coals. | Per-Ortho-Bituminous. Cannels. | Per-Para-Bituminous. | Per-Lignitous | .... |
| H over 5·8 .. | .... | .... | Over 5·7 | Over 5·7 | Over 5·8 | Over 5·8 | .... |
| Vol. 30 to 44 .. | .... | .... | 30 to 44 | Seldom under 36 | Seldom under 40 | 31 to 57 | .... |
| Bituminous .. | .... | Pseudo-Bituminous | Meta-Bituminous. Welsh coking coals. | Ortho-Bituminous. Durham coking coals. Smithy and gas coals. | Para-Bituminous. Gas and coking and free-burning steam coals. | Lignitous | .... |
| H 5·0 to 5·8 .. | .... | .... | .... | .... | 5·0 to 5·8 | 4·7 to 5·8 | .... |
| Vol. 23 to 30 .. | .... | .... | .... | .... | 30 to 40 | 31 to 57 | .... |
| Semi-Bituminous | .... | Semi-Bituminous. Generally coking steam coals. | Sub-Meta-Bituminous. Continental coking coals. | Sub-Ortho-Bituminous. Westphalian coking coals. | Sub-Para-Bituminous. | Sub-Lignitous | .... |
| H 4·5 to 5·0 .. | .... | Over 4·45 | 4·5 to 4·9 | 4·5 to 5·0 | 5·0 and under | .... | .... |
| Vol. 16 to 23 .. | .... | 14 to 24 | 16 to 23 | 16 to 23 | 16 to 29 | .... | .... |
| Carbonaceous .. | Semi-Anthracitic. Non-coking, dry steam coals. | Ortho-Carbonaceous. Welsh smokeless, slightly coking coals. | Pseudo-Carbonaceous. (Sub-Meta-Bituminous) steam coals. | Pseudo-Carbonaceous. (Sub-Ortho-Bituminous). | Pseudo-Carbonaceous. (Sub-Para-Bituminous). | Sub-Lignitous | .... |
| H 4·0 to 4·5 .. | Over 4 | 4·2 to 4·5 | 3·7 to 4·5 | Over 4·5 | .... | .... | .... |
| Vol. 10 to 16 .. | 9 to 15 | 10 to 14 | 10 to 16 | Over 16 | .... | .... | .... |
| Anthracitic .. | Ortho-Anthracite. True anthracites. | Pseudo-anthracite (Sub-Carbonaceous). Dry Steam coals. Bastard anthracites. | Pseudo-anthracitic (Sub-Meta-Bituminous). | Pseudo-anthracitic (Sub-Ortho-Bituminous). | Pseudo-anthracitic (Sub-Para-Bituminous). | .... .... | .... .... |
| H below 4·0 .. | Below 4·0 | Below 4·2 | Below 3·7 | .... | .... | .... | .... |
| Vol. under 10 .. | 5 to 9 | Under 7·7 | Under 10 | .... | .... | .... | .... |



COALFIELDS AND LIGNITE OCCURRENCES OF INDIA
500 0 500 km
LIGNITE
LIGNITE
DAMODAR-KOEL VALLEY COALFIELDS
SON-MAHANADI VALLEY COALFIELDS
GODAVARI VALLEY COALFIELDS
Srinagar
Simla
Chandigarh
New Delhi
Jaipur
Lucknow
Patna
Gangtok
Itanagar
Dispur
Shillong
Kohima
Imphal
Agartala
Aizawl
Calcutta
Gandhinagar
Bhopal
Bhubaneswar
Mumbai
Hyderabad
Bangalore
Chennai
Trivandrum
68° 72° 76° 80° 84° 88° 92° 96°
36° 32° 28° 24° 20° 16° 12° 8°
Lignite Occurrences
1. Nichahom Baramula
2. Nagbal
3,4. Firozpur Shaliganga
5. Khari
6. Gangashahr
7,8. Sarup Desar
9. Kolayat
10. Gura
11. Chaneri
12. Bithnok
13. Mandal Charnan
14. Raneri
15. Hira Ki Dhani
16. Lalamdesar
17. Bar Singhsar
18. Palana
19. Kasnau
20. Kuchera
21. Nimbri
22. Marwar Chapri
23. Butati
24. Merta Road
25. Hansiyas
26. Gangarda
27. Lamba Jatan
28. Mokala
29. Indawar
30. Nagarda
31. Nimla
32. Bharka
33. Botiya
34. Hamirji ki Dhani
35. Kapurdi
36. Jalipa
37. Lakhpat
38. Dhedadi
39. Pranpur
40. Kaiyari
41. Panandhro
42. Akhrimota
43. Fulradam
44. Umarsar
45. Mudia
46. Mata No Madh
47. Jularai
48. Waghapadar
49. Lefri
50. Hamla
51. Nanaratadia
52. Lakhanka Kharsalia
53. Bhuri
54. Shah Vastan
55. Ratnagiri
56. Nileswaram
57. Madayi
58. Warkala
59. Mannargudi
60. Jayamkonda Cholapuram
61. Lalpettai
62. Srimushnam
63. Neyveli South
64. Bhubangiri
65. Neyveli East
67,68,69. Eluru - Rajamun
70. Domra
Gondwana Coalfields
22. Singrimari
23. Abor Hills
24. Daphla Hills
25. Aka Hills
26. Darjeeling
27. Rajmahal
28. Birbhum
29. Tangsuli
30. Kundait Kariah
31. Saharjuri
32. Jainti
33-34. Giridih
35. Raniganj
36. Barjora
37. Jharia
38. East Bokaro
39. West Bokaro
40. Ramgarh
41. North Karanpura
42. South Karanpura
43. Chope
44. Itkhori
45. Auranga
46. Hutar
47. Daltonganj
48-49-50. Tatapani-Ramkola
51. Singrauli
52. Korar
53. Umaria
54. Johilla
55. Sohagpur
56. Sonhat-Jhilimili
57. Bisrampur
58. Lakhanpur
59. Panchbahini
60. Dhamamunda
61. Koreagarh
62. Chirimiri
63. Sendurgarh
64. Hasdo-Arand
65. Korba
66. Mand-Raigarh
67. Ib River
68. Talcher
69. Godavari
70. Yellendu
71. Wardha
72. Bander
73. Umrer
74. Kamptee
75. Pench Kanhan
76. Pathakhera
77. Tawa
78. Mohpani
Tertiary Coalfields
1. Miao-bum
2. Namchik - Namphuk
3. Makum
4. Dilli - Jeypore
5. Tiru
6. Borjan
7. Tuen Sang
8. Jhanzi-Disai
9. Koilajan
10. Diphu
11. Bapung
12. Pynursia
13. Cherrapunji
14. Laitryngew
15. Mawsynram
16. Mawlong Shella
17. Langrin
18. Balphakram - Pendenggru
19. West Daranggiri
20. Rongrenggri
21. Siju
79. Kalakot - Mahogala

# GEOLOGICAL AGE OF INDIAN COAL DEPOSIT

- **OCCURRENCE OF COAL AND LIGNITE DEPOSIT**

| GEOLOGICAL AGE (IN MILLION YRS) | GEOLOGICAL FORMATION/GROUP | OCCURRENCE OF COAL/LIGNITE |
|---|---|---|
| **TERTIARY** | | |
| PLEISTOCENE (1) | KAREWAS | KASHMIR LIGNITE |
| UPPER-MIOCENE-PLIOCENE(25) | CUDDALORE BEDS | SOUTH ARCOT, NEYVELI LIGNITE TAMIL NADU |
| OLIGOCENE (38) | TIKAK PARBAT FORMATION OF BARAIL GROUP | COAL OF UPPER ASSAM,ARANUCHAL PRADESH AND NAGALAND |
| EOCENE (65) | LAKI AND JAINTIA GROUP | LIGNITES OF RAJASTHAN AND GUJRAT,COALS OF J&K,LOWER ASSAM AND MEGHALAYA |
| **GONDWANA** MESOZOIC | | |
| LOWER CRETACEOUS (140) | UMIA, JABALPUR | THIN COAL SEAMS OF GUJRAT |
| LOWER JURASSIC | KOTA AND CHIKIALA FORMATION | THIN COAL SEAMS OF SATPURA AND GODAVARI |
| PALEOZOIC | | |
| UPPER PERMIAN (225) | RANIGANJ FORMATION AND ITS EQUIVALENT | LOWER GONDWANA C.F. OF PENINSULAR INDIA AND FOOT HILLS REGION OF EASTERN HIMALAYAD |
| MIDDLE PERMIAN (270) | BARAKAR FORMATION | ” |
| LOWER PERMIAN (290) | KARHARBARI FORMATION | ” |

# GONDWANA COAL

## Coal bearing strata

***RANIGANJ/KAMTHI FORMATION*** **(Late Permian 245-258 Ma)–**

**ECONOMICALLY EXPLOITABLE IN RANIGANJ, JHARIA EASTERN PART OF SINGRAULI BASIN AND GODAVARI**

***BARAKAR FORMATION*** **(Early Permian 258-286 Ma)–**

**MAJOR STOREHOUSE OF COAL IN ALL THE BASINS**

***KARHARBARI FORMATION*** **(Early Permian 258-286 Ma)–**

**RESTRICTED TO FEW COALFIELDS OF EASTERN INDIA**

# TERTIARY COAL

## Coal bearing strata

- **Oligocene sediments (23-36 Ma)–**
  - **Tikak Parbat Formation in Upper Assam, Nagaland and Arunachal Pradesh**
- **Eocene sediments (36-57 Ma)– Tura Sandstone, Lakadong Sandstone in Garo, Khasi and Jaintia hills of Meghalaya**
  - **Sylhet Limestone in Mikir hills of Assam**
  - **Lower Subathu Group in Jammu**

## Gondwana coal

Occurrence Eastern and central part of Peninsular India

Rank Bituminous to sub-bituminous

Character Moderate to High in Ash and Low in Sulphur

## Tertiary coal

Occurrence Northeastern India

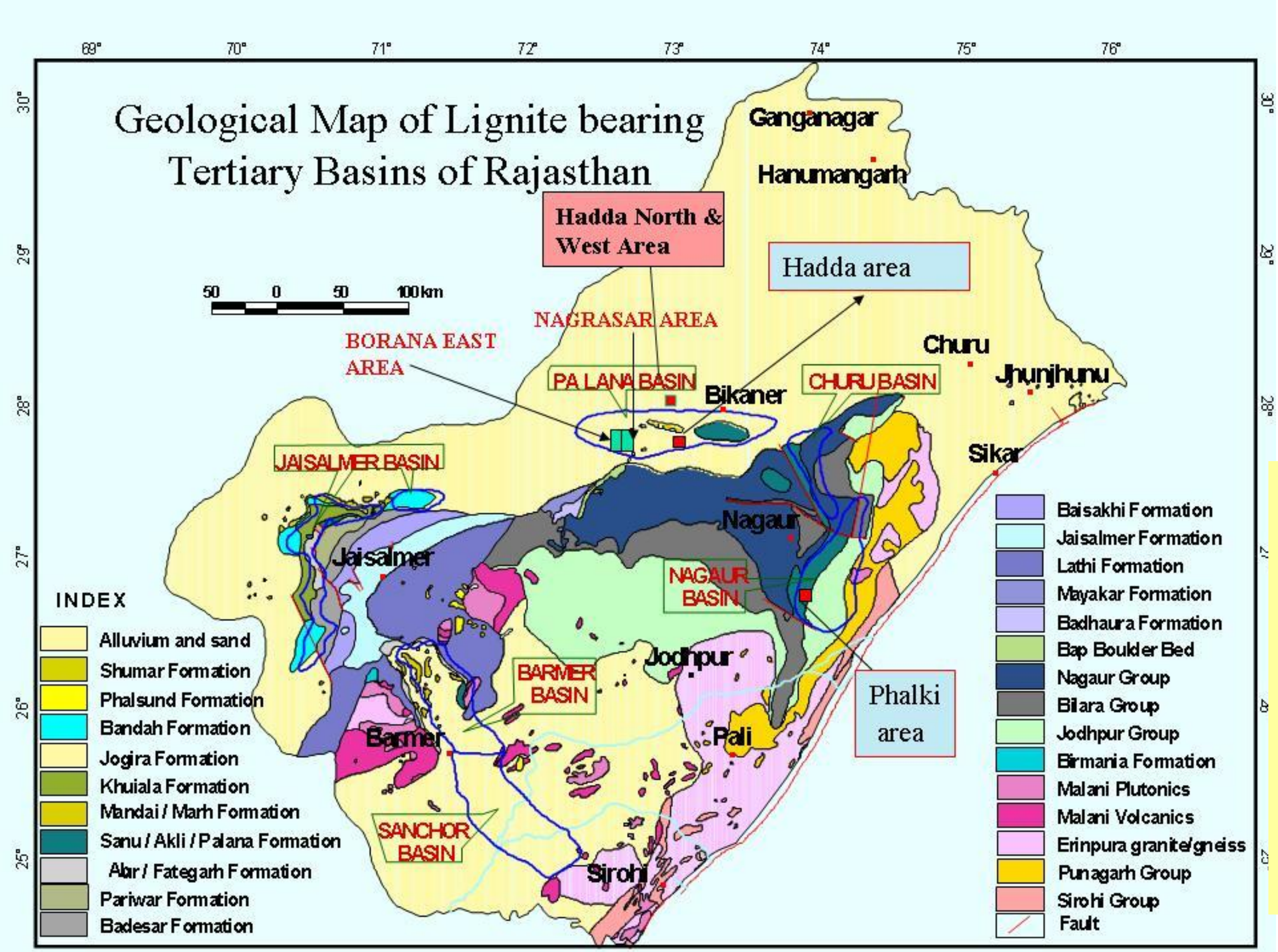
Rank Meta and Ortholignitous

Character High in Sulphur ; Strongly caking to non-caking

## Lignite

Occurrence Southern and Western India

Character High in Moisture and Volatile Matter




# RAJASTHAN

**Total Resource:**

**4554 m.t.**

**Lignite seams are generally thin and development is mostly lenticular and discontinuous**

**Lignite occurs mainly in 3 basins.**

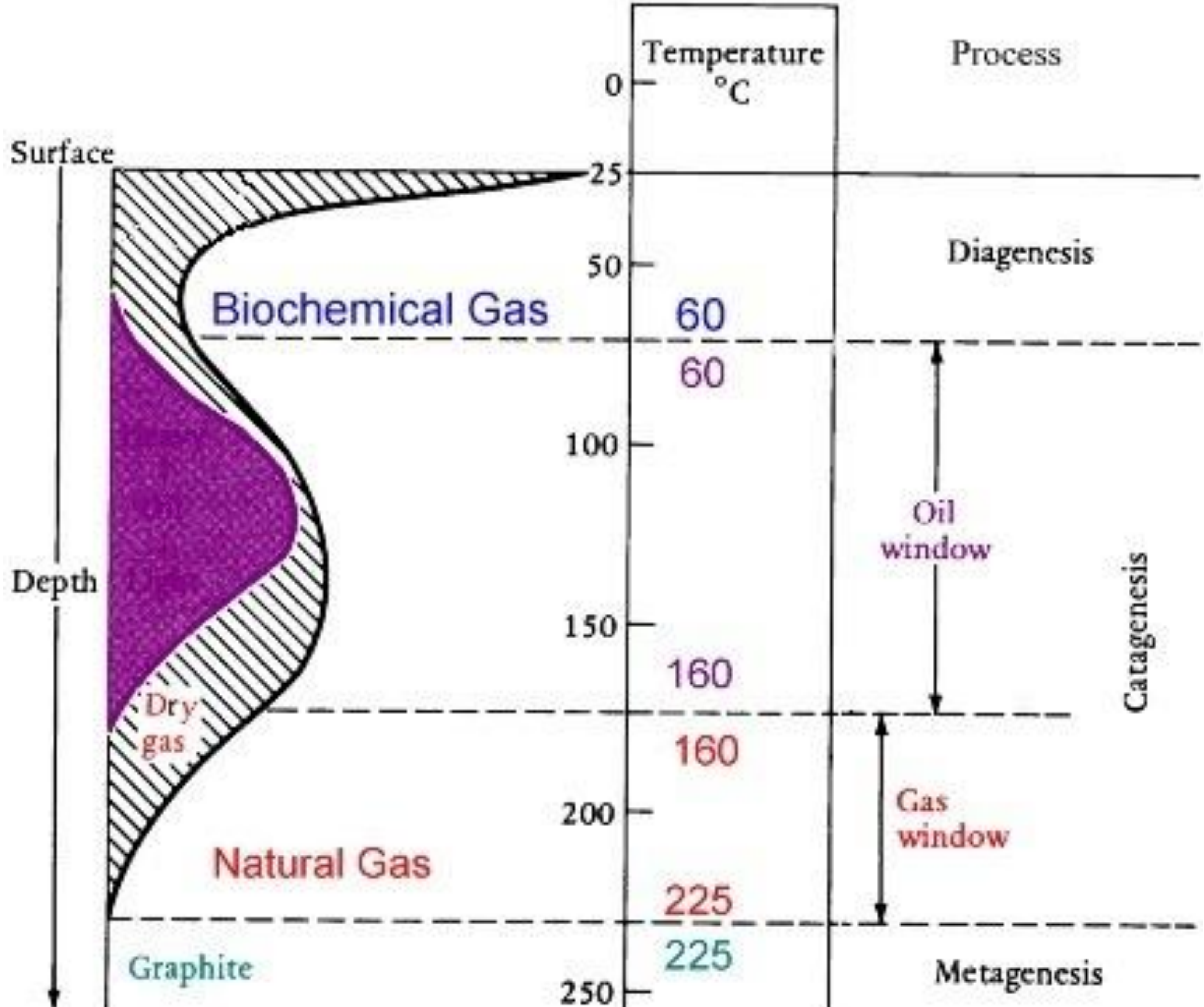
**Barmer : 3422 m.t.**

**Bikaner : 809 m.t**

**Nagaur : 233.42**

# HYDROCARBON RESOURCES OF INDIA

- **Natural Gas**
- **Natural gas** (also called fossil gas; sometimes just gas) is a **naturally occurring hydrocarbon gas mixture consisting primarily of methane**, but commonly including varying amounts of other higher alkanes, and sometimes a small percentage of carbon dioxide, nitrogen, hydrogen sulfide, or helium.
- Natural Gas is also a fossil fuel and is **found in the same geological structure where petroleum is found.**
- Sometimes, the pressure of natural gas forces oils up to the surface. Such natural gas is known as **associated gas or wet gas.**
- Some reservoirs contain gas and **no oil**. This gas is termed **non-associated gas or dry gas.**
- Often natural gases contain substantial quantities of **hydrogen sulfide** or other organic sulfur compounds. In this case, the gas is known as **"sour gas."**
- Coalbed methane is called **'sweet gas'** because of its lack of hydrogen sulfide.
- On the market, natural gas is usually bought and sold not by volume but by **calorific value.**
- In practice, purchases of natural gas are usually denoted as MMBTUs (millions of British thermal units (BTU or Btu)) = ~1,000 cubic feet of natural gas.
- **Natural gas is odorless and colorless**; the slightly sour smell that we associate with the gas coming from a stovetop is due to an odorization process (for safety and leak detection) which adds mercaptan compounds to the end-use gas.



Temperature
°C
Process
0
Surface
25
Diagenesis
50
Biochemical Gas
60
60
100
Oil
window
Depth
Catagenesis
150
160
Dry
gas
160
Gas
window
200
Natural Gas
225
225
Graphite
250
Metagenesis

**Diagenesis**

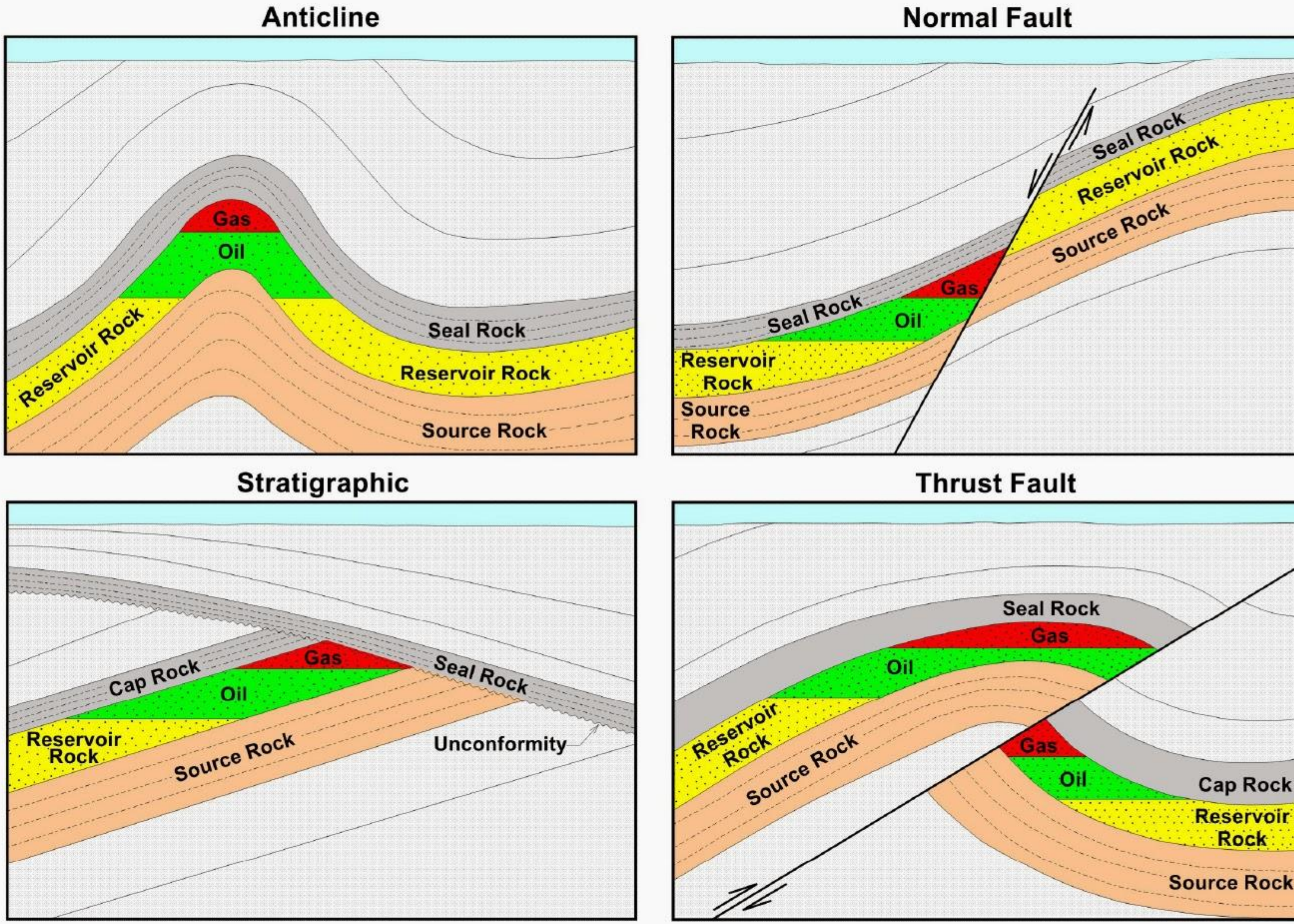
- ′′**Diagenesis** is defined as the chemical reactions that occurred in the first few thousand years after burial at temperatures less than 50◦C. During diagenesis: oxygen, nitrogen and sulfur are removed from the organic matter leading to an increase in hydrogen content of the sedimentary organic matter. In addition, presence of iron during diagenesis increases the probability of sulfur removal as the iron reacts with sulfur over long burial time and this increases petroleum economic value.

**Catagenesis**

- ′′The reactions that occur between 60 and 200◦C are considered to be catagenetic in nature. During catagenesis, the organic compounds are exposed to diverse thermal degradation reactions that include double bonds reduction by adding sulfur or hydrogen atoms, cracking reactions and condensation reactions.

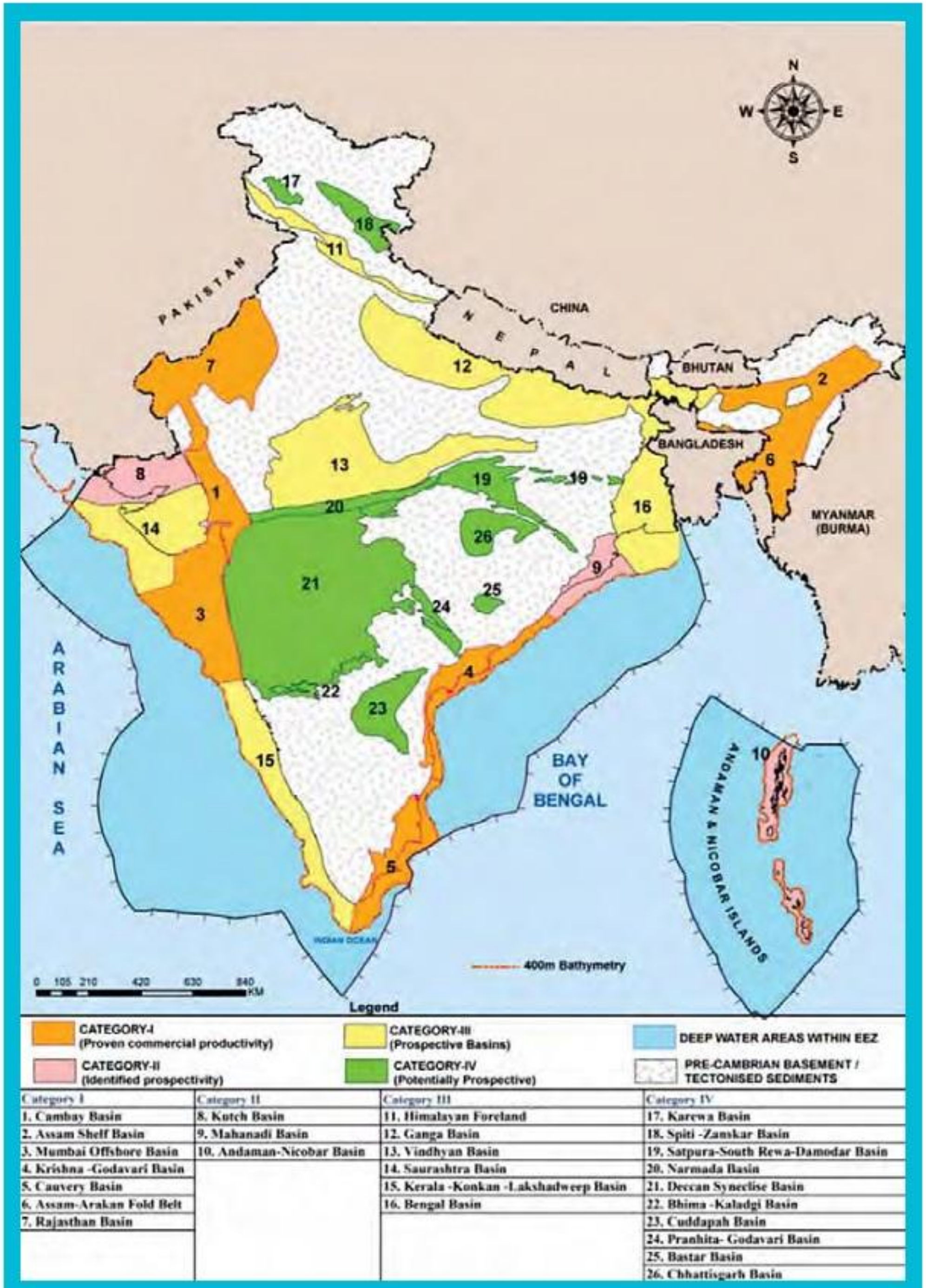
**Metagenesis**

- ′′**Metagenesis** takes place at temperatures over 200◦C and is considered to be a type of very low-grade metamorphism.




Simplified cartoons showing some common types of hydrocarbon traps recognized in producing oil and gas fields around the world.

# Sedimentary Basins of India




26 Sedimentary Basins have have been classified into four categories as below

**Category-I:** Basins with **established commercial production**. Cambay, Mumbai Offshore, Rajasthan, Krishna Godavari, Cauvery, Assam Shelf and Assam-Arakan Fold Belt

**Category-II:** Basins with known accumulation of hydrocarbons but no. commercial production achieved so far: Kutch, Mahanadi-NEC (North East Coast) Basin, Andaman-Nicobar, Kerala-Konkan-Lakshadweep Basin.

**Category-III:** Basins are considered geologically prospective Himalayan Foreland Basin, Ganga Basin, Vindhyan basin, Saurashtra Basin, Kerela Konkan Basin, Bengal Basin

**Category-IV:** Basins having uncertain potential which may be prospective by analogy with similar basins in the world. Karewa basin, Spiti-Zanskar basin, Satpura–South Rewa–Damodar basin, Chhattisgarh Basin, Narmada basin, Deccan Syneclise, Bhima-Kaladgi, PranhitaGodavari.